THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 8: **Introduction to Trigonometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $\sin A = \frac{5}{13}$, then $\cos A$ is: 1

(a) $\frac{12}{13}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{13}{12}$ **(d)** $\frac{8}{13}$

2. The value of $\sin 30^\circ + \cos 60^\circ$ is: 1

(a) $\frac{1}{2}$ **(b)** 1 **(c)** $\frac{\sqrt{3}}{2}$ **(d)** 0

3. $\frac{1-\tan^2 30^\circ}{1+\tan^2 30^\circ}$ equals: 1

(a) $\sin 60^\circ$ **(b)** $\cos 60^\circ$ **(c)** $\tan 60^\circ$ **(d)** $\tan 30^\circ$

4. If $\tan A = 1$, then $2\sin A \cos A$ is: 1

(a) 0 **(b)** 1 **(c)** $\sqrt{2}$ **(d)** $\frac{1}{2}$

5. A student states: "As θ increases from 0° to 90°, $\cos\theta$ increases." The correct response is: 1

(a) Correct **(b)** Incorrect; $\cos\theta$ decreases from 1 to 0

(c) Correct up to 45° **(d)** Cannot be determined

6. $\sec A(1 - \sin A)(\sec A + \tan A)$ equals: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** $\sec A$

7. Student A says $\tan 60^\circ = \sqrt{3}$; Student B says $\tan 60^\circ = \frac{1}{\sqrt{3}}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. In $\triangle ABC$, right-angled at B, $AB = 5$ and $AC = 13$. Then $\tan C$ is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{12}{5}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{12}{13}$

9. $9\sec^2 A - 9\tan^2 A$ equals: 1

(a) 1 **(b)** 9 **(c)** 8 **(d)** 0

10. If $2\sin\theta = 1$ (θ acute), then θ is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Given $\sec\theta = \frac{13}{12}$, calculate all the other trigonometric ratios. 2

12. Evaluate: $\frac{\sin 30^\circ + \tan 45^\circ - \csc 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$. 2

13. If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, \$0^\circB, $findAandB$\$. 2

14. A student says that $\sin^2\theta + \cos^2\theta = 1$ is true only for $\theta = 45°$. Refute the claim. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $4\tan\theta = 3$, evaluate $\frac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1}$. 3

16. Prove that $\frac{1+\sec A}{\sec A} = \frac{\sin^2 A}{1-\cos A}$. 3

17. Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$. 3

OR

Prove that $\frac{1+\tan^2 A}{1+\cot^2 A} = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$. 3

18. In $\triangle ABC$, right-angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, find the value of $\sin A\cos C + \cos A\sin C$. 3

OR

Express $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that $\sqrt{\frac{\sec\theta - 1}{\sec\theta + 1}} + \sqrt{\frac{\sec\theta + 1}{\sec\theta - 1}} = 2\csc\theta$. 5

OR

Prove that $(\csc A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$. 5

20. **Case study:** A carpenter's set-square is a right triangle PQR, right-angled at Q, with $PQ = 8$ cm and $QR = 6$ cm. 5

(i) Find PR.

(ii) Find $\sin P$ and $\cos R$.

(iii) Why is $\sin P = \cos R$?

(iv) Show that $\tan P \cdot \tan R = 1$.

OR

If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, show that $m^2 - n^2 = 4\sqrt{mn}$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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